

ECSE-4040

Digital Electronics

Lecture 25

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Lecture Outline

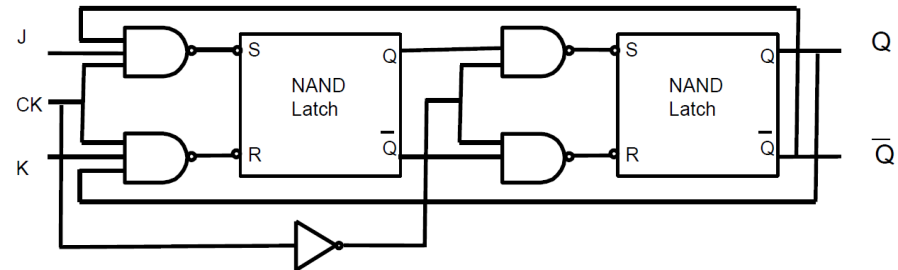
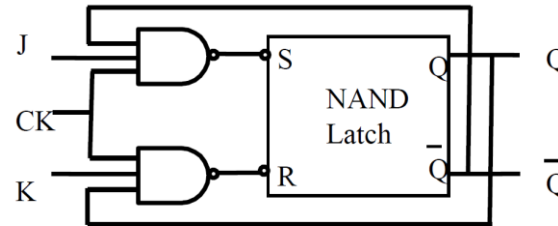
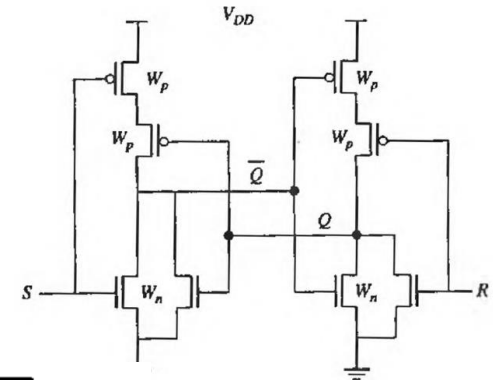
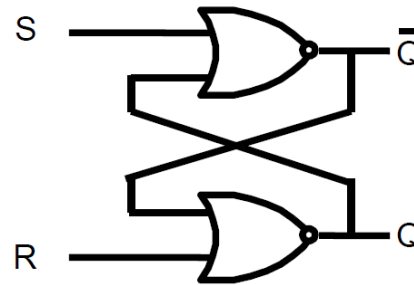
- SRAM
 - Memory Array
 - SRAM Cell

RAM Overview

- Need to store data created by logic
- This data can be randomly accessed at any time, dependent on program requirements.
- Two families of RAM
 - Static (SRAM)
 - Storage element created by dual inverter loop.
 - 6 to 8 transistors (4 for basic storage element)
 - Dynamic (DRAM)
 - Storage element is a MOS capacitor (NMOS without S or D connections)
 - 1 transistor total

Why not latches, flip-flops?

- Basic NAND/NOR latch
 - 8 Transistors
- Level-triggered JK flip-flop
 - 20 Transistors
 - 22 for D-Flip-Flop
- Edge-triggered JK flip-flop
 - 38 Transistors
 - *There are hacks for 11T DFF
- More transistors = bigger RAM cell
- Bigger RAM cell = less RAM
- Conclusion – RAM arrays are very simple, small storage cells



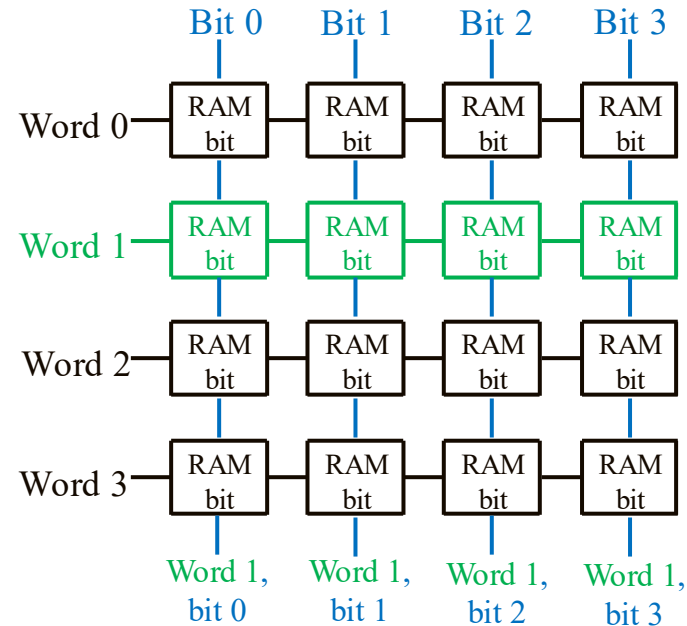
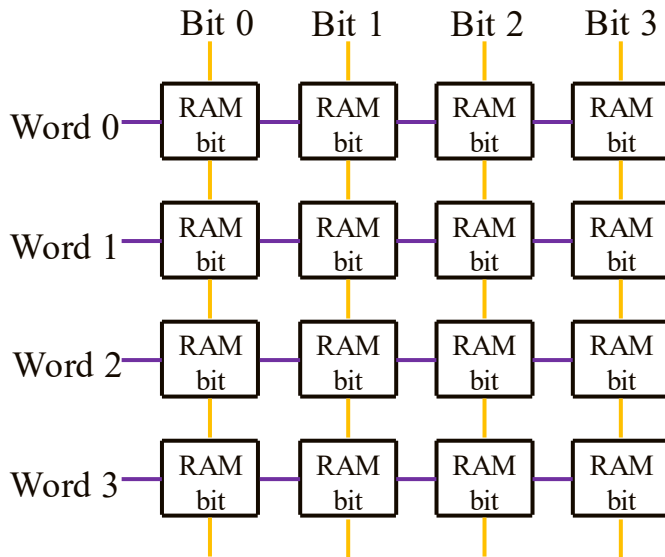
SRAM

- Static
 - Memory data is stored indefinitely as long as power is maintained
 - Non-destructive read-out
 - Do not need to re-write data after read
- Lower logic density
 - 6 transistors
 - 2 bit-lines per cell
- Faster
 - Inverter provides continual current drive on output bit-lines

DRAM

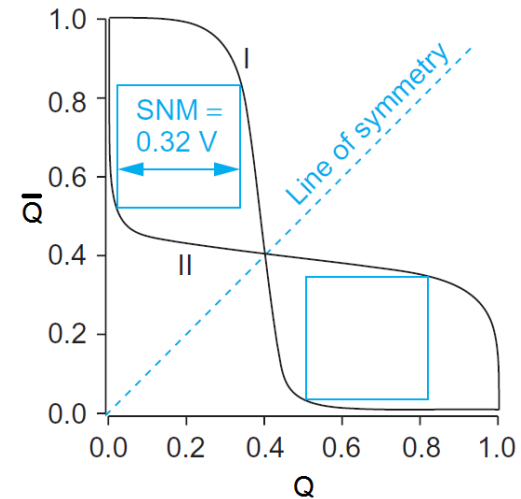
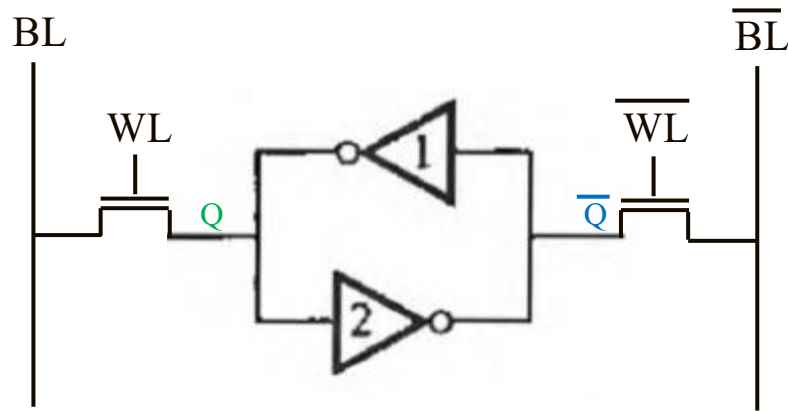
- Dynamic
 - memory data is lost over time even if power is maintained
 - Capacitors slowly drain
 - Must be refreshed every $\sim 10\mu\text{s}$
 - Destructive read-out
 - Must re-write data back
- Higher logic density
 - Just 1 transistor
 - Capacitor is stacked below transistor
 - 1 bit line, word line per cell
- Slower
 - Capacitor only provides a small burst of charge to/from the bit-line

RAM Arrays

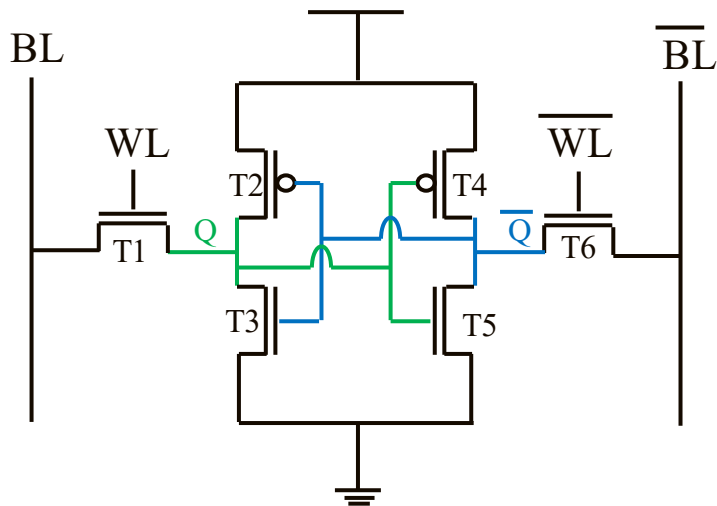
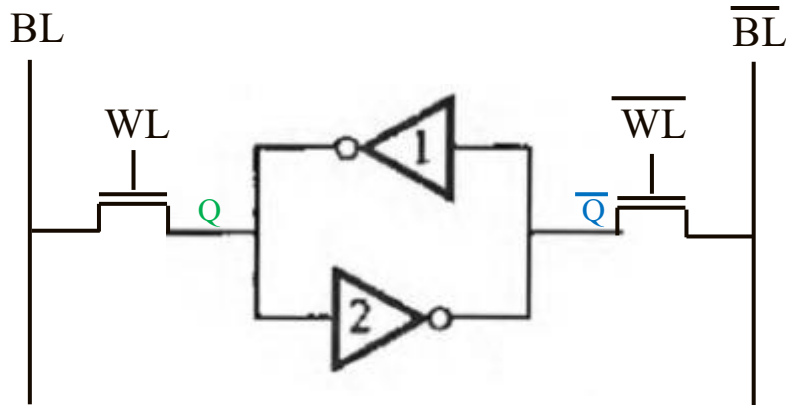


- Ram is organized in 2-D grid
 - Horizontal D is Word (memory address)
 - Vertical D is Bit
 - This is a 4x4 memory
 - 4 words by 4 bits
- Word Line (WL) activates row
 - Connects each memory bit in the word to its respective bit line
 - Read – memory bit sets entire bit line
 - Write – bit line sets the activated memory bits

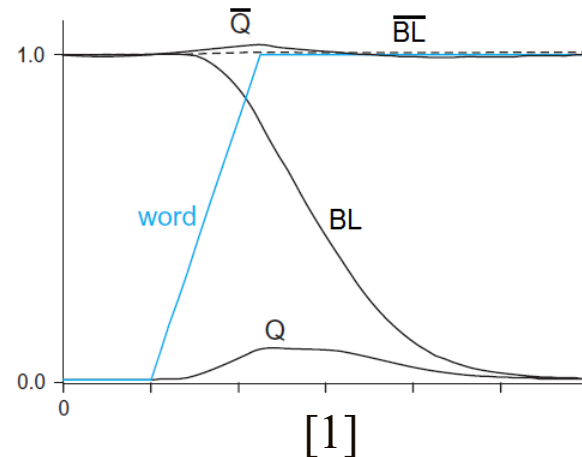
SRAM Cell



SRAM - Read

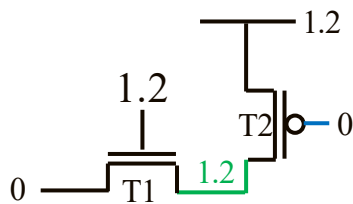


- Bit-line access controlled by NMOS pass transistor
 - Degraded, slow logic 1 pull-up
- One side (BL, WL) is read port
- Other side ($\overline{BL}$, $\overline{WL}$) is write port
- To Read
 - Set WL high
 - Q inverter pulls BL to full logic 0 or 1
 - T2,T3 must overpower T1, or stored data is destroyed
 - $W_{T2,T3} > W_{T1}$

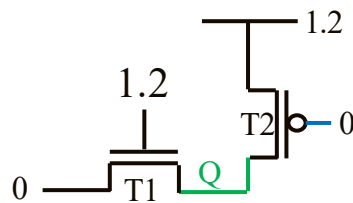


SRAM - Read

- Example – read 1
- Initial condition



- Ensure Q does not drop below V_{IH}



$$W_{T2} = 5W_{T1}$$

$$L = 100\text{nm}$$

Assume T2 linear, T1 saturated

$$\mu_P \frac{W_P}{L} \cancel{\text{ox}} \left[(V_{SG} - |V_{TP}|) (V_{DD} - V_Q) - \frac{(V_{DD} - V_Q)^2}{2} \right] \frac{1}{1 + \frac{V_{DD} - V_Q}{E_c L}}$$

$$= V_{sat} \frac{W_N \cancel{\text{ox}} (V_{GS} - V_{TN})^2}{V_{GS} - V_{TN} + E_c L}$$

$$\frac{70.5 \cdot}{1 + \frac{1.2 - V_Q}{2.4}} \left[(1.2 - 0.4)(1.2 - V_Q) - \frac{(1.2 - V_Q)^2}{2} \right]$$

$$= \frac{8 \cdot 10^6 \cdot 10^{-9} \cdot (1.2 - 0.4)^2}{1.2 - 0.4 + 0.6}$$

$$V_Q = 1.05\text{V}$$

SRAM - Read

$$V_a = 1.05V$$

NMOS check

$$V_{DSN} = 1.05V \quad \text{Sat if } V_{DS} \geq V_{GS} - V_T$$
$$1.05 \geq 1.2 - 0.4$$
$$1.05 \geq 0.8 \quad \checkmark$$

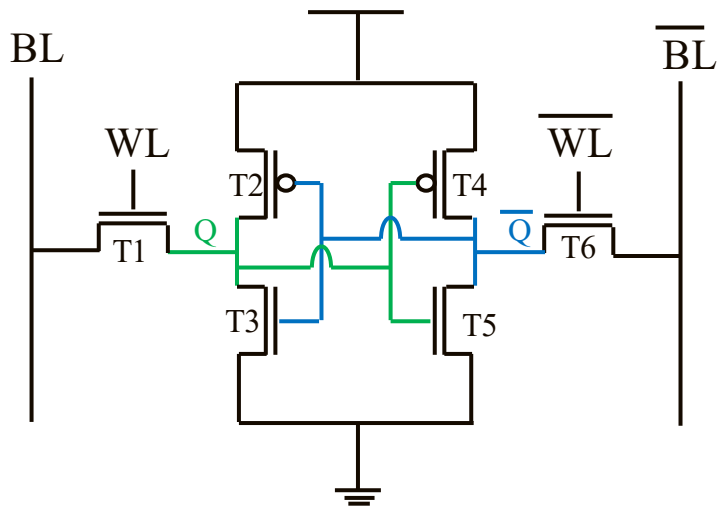
PMOS check

$$V_{SDP} = 1.2 - 1.05$$
$$V_{SDP} = 0.15V$$
$$V_{SD}(\text{sat}) = \frac{(V_{SG} - |V_{TP}|)E_{cPL}}{V_{SG} - |V_{TP}| + E_{cPL}} = \frac{(1.2 - 0.4)2.4}{1.2 - 0.4 + 2.4}$$
$$V_{SD}(\text{sat}) = 0.6V$$

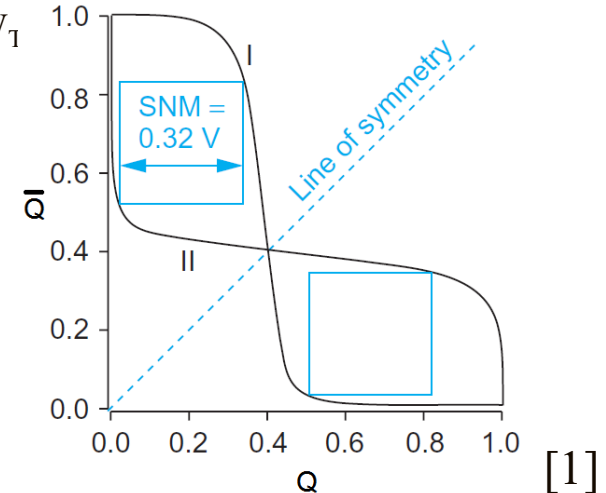
Linear if

$$V_{SDP} \leq V_{SD}(\text{sat})$$
$$0.15V \leq 0.6V \quad \checkmark$$

SRAM - Write

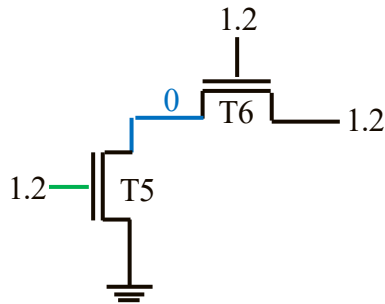


- One side (BL, WL) is read port, other side ($\overline{\text{BL}}$, $\overline{\text{WL}}$) is write port
- To write
 - Set both $\overline{\text{WL}}$ and WL high
 - $\overline{\text{BL}}$ overpowers T4, T5 inverter to pull $\overline{\text{Q}}$ to $\overline{\text{BL}}$ voltage
- Remember $W_{T2,T3} > W_{T1}$ to support read operations on BL, WL side
- For write, T1 must overpower T2, T3, or state cannot change
 - Not possible since port is designed NOT to lose state on read
 - Design T4, T5, T6 for write
 - T6 must overpower T4, T5
 - $W_{T6} > W_{T1}$



SRAM - Write

- Example – write 1
- Initial condition



With W_{T5} only $2 \times W_{T6}$, this will be close.

Estimate: $1.2 \frac{W_6}{W_5 + W_6} = 0.8$, $V_{DS} > 0.4V$, assume both saturated

$$I_{DSS} = I_{DS6}$$

$$\frac{\cancel{V_{sat}} \cancel{W_{T5}} \cancel{C_{ox}} (V_{GSS} - V_{TN})^2}{V_{GSS} - V_{TN} + E_c L} = \frac{\cancel{V_{sat}} \cancel{W_{T6}} \cancel{C_{ox}} (V_{GSS} - V_{TN})^2}{V_{GSS} - V_{TN} + E_c L}$$

$$\frac{(1.2 - 0.4)^2}{1.2 - 0.4 + 0.6} = \frac{2 \cdot (1.2 - V_a - 0.4)^2}{1.2 - V_a - 0.4 + 0.6}$$

$$V_a = 0.3V \text{ fails!}$$

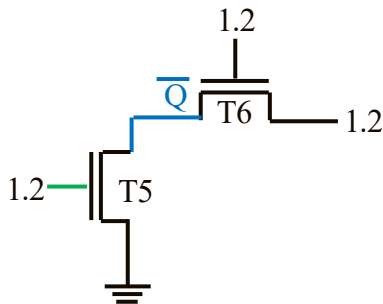
$$V_{IH} \approx 0.7V$$

For success, $W_{T6} \approx 20W_{T5}$

$$W_{T6} = 2W_{T5}$$

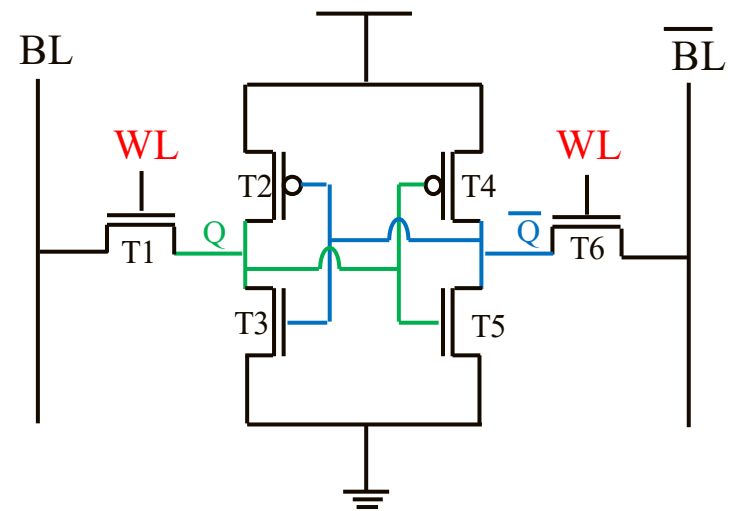
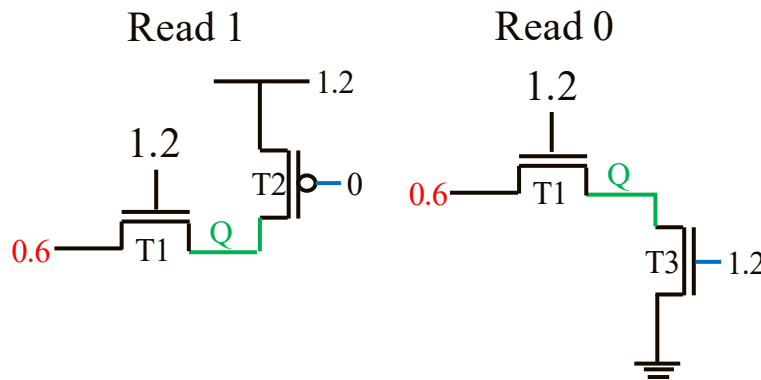
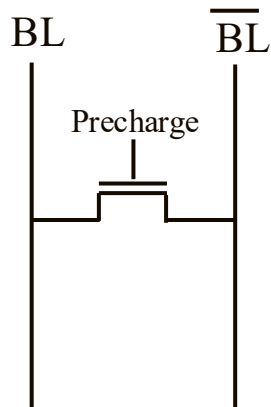
$$L = 100nm$$

- Ensure $\bar{Q}$ does exceed V_{IH}



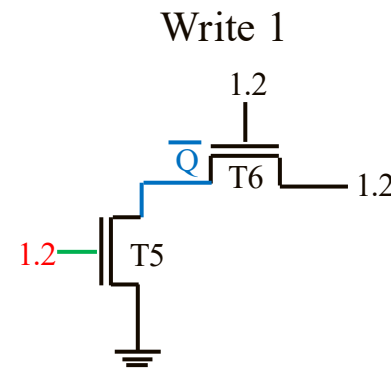
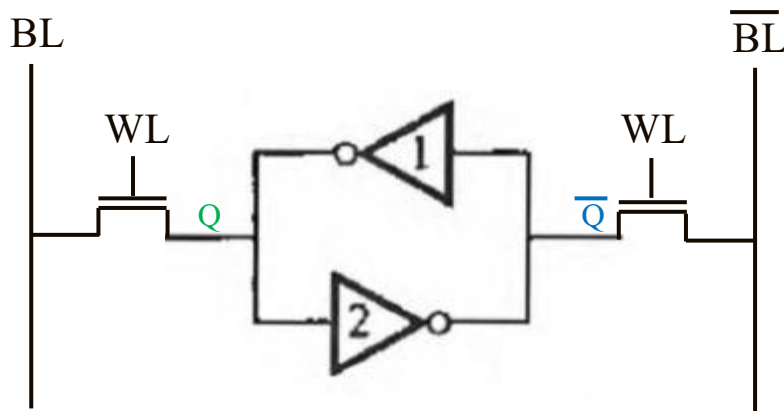
Pre-charging

- As seen previously, for write, excessively large W_{T6} is required
 - Read port encounters same issue for read 0
- Solution – Precharging
- Set both BL and $\overline{BL}$ to $V_{DD}/2$ prior to read
- Can now eliminate $\overline{WL}$, balance transistor sizing



Pre-charging

- For write, second inverter now assists substantially
- Opposing transistor will now begin to turn off prior to reaching V_{IH}



V_{GS} now declines when V_{IL} is surpassed on Q (SRAM is now symmetric and changes in Q and $\overline{Q}$ mirror each other).